

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 6

4 Computation Theory (amp12)

Lambda calculus (a) For the λ -calculus, define the notions of

(i) β -conversion ($=_\beta$) [2 marks]

Answer: β -conversion is the smallest binary relation between λ -terms that is a congruence and contains all (redex, reduct)-pairs, $((\lambda x.M)N, M[N/x])$.

(ii) Church numeral ($\underline{n}$) [2 marks]

Answer: The n th Church numeral is $\underline{n} \triangleq \lambda f x. f^n x$, where in general

$$M^n N \triangleq \begin{cases} N & \text{if } n = 0 \\ M(M^n N) & \text{if } n = m + 1 \end{cases}$$

(b) What does it mean for a total function $f : \mathbb{N}^n \rightarrow \mathbb{N}$ to be λ -definable? Explain why it is the case that not every $f : \mathbb{N}^n \rightarrow \mathbb{N}$ is λ -definable, carefully stating any standard results that you rely upon. [3 marks]

Answer: $f : \mathbb{N}^n \rightarrow \mathbb{N}$ is λ -definable if there is a closed λ -term F such that for all $(x_1, \dots, x_n) \in \mathbb{N}^n$, $F \underline{x_1} \dots \underline{x_n} =_\beta f(x_1, \dots, x_n)$. [No points deducted if the candidate gives the more complicated definition for *partial* functions. The candidate needs to say that F is closed.]

It is a standard result that the class of λ -definable partial numerical functions coincides with the class of computable partial functions; so every λ -definable $f : \mathbb{N}^n \rightarrow \mathbb{N}$ has to be computable. And of course not every function is computable (another standard result).

(c) Explain why the predecessor function $\text{pred} : \mathbb{N} \rightarrow \mathbb{N}$

$$\text{pred}(x) = \begin{cases} 0 & \text{if } x = 0 \\ x - 1 & \text{if } x > 0 \end{cases}$$

is λ -definable and give a λ -term that represents it. [4 marks]

Answer: Given $f : \mathbb{N} \rightarrow \mathbb{N}$, iterating the function

$$g_f : (x, y) \mapsto (f(x), x)$$

$n + 1$ times starting from (x, x) gives the pair $(f^{n+1}(x), f^n(x))$. So we can get $f^n(x)$ from $f^{n+1}(x)$ parametrically in f and x , by building g_f from f , iterating $n + 1$ times from (x, x) and then taking the second component. Hence

$$\text{Pred} \triangleq \lambda y f x. \text{Snd}(y (G f)(\text{Pair } x x))$$

— *Solution notes* —

has the required β -reduction properties to λ -define *pred*, where

$$\begin{aligned} G &\triangleq \lambda f p. \text{Pair}(f(\text{Fst } p))(\text{Fst } p) \\ \text{Pair} &\triangleq \lambda x y f. f x y \\ \text{Fst} &\triangleq \lambda f. f \text{ True} \\ \text{Snd} &\triangleq \lambda f. f \text{ False} \\ \text{True} &\triangleq \lambda x y. x \\ \text{False} &\triangleq \lambda x y. y \end{aligned}$$

(d) Show that the following functions are λ -definable. For each part you may assume solutions to the previous parts of the question.

(i) $if_0 : \mathbb{N}^3 \rightarrow \mathbb{N}$, where $if_0(x, y, z) = \begin{cases} y & \text{if } x = 0 \\ z & \text{if } x \neq 0 \end{cases}$ [3 marks]

Answer: if_0 is λ -defined by the closed term $\text{Eq}_0 \triangleq \lambda x.x(\lambda z.\text{False}) \text{ True}$ [given in lectures], where **True** and **False** are as in the solution to part (c). For then we have $if_0 \underline{x} \underline{y} \underline{z} =_{\beta} (\lambda z. \text{False})^x \text{True} \underline{y} \underline{z}$ and hence

$$\text{Eq}_0 \underline{0} \underline{y} \underline{z} =_{\beta} \text{True} \underline{y} \underline{z} =_{\beta} \underline{y} = \underline{if_0(0, y, z)}$$

and

$$\text{Eq}_0 \underline{x+1} \underline{y} \underline{z} =_{\beta} (\lambda z. \text{False})((\lambda z. \text{False})^x \text{True}) \underline{y} \underline{z} =_{\beta} \text{False} \underline{y} \underline{z} =_{\beta} \underline{z} = \underline{if_0(x+1, y, z)}$$

(ii) $and : \mathbb{N}^2 \rightarrow \mathbb{N}$, where $and(x, y) = \begin{cases} 0 & \text{if } x = 0 \text{ and } y = 0 \\ 1 & \text{if } x \neq 0 \text{ or } y \neq 0 \end{cases}$ [1 mark]

Answer: It follows from part (d)(i) that *and* is λ -definable by

$$\text{And} \triangleq \lambda x y. \text{Eq}_0 x (\text{Eq}_0 y \underline{0} \underline{1}) \underline{1}$$

(iii) $monus : \mathbb{N}^2 \rightarrow \mathbb{N}$, where $monus(x, y) = \begin{cases} x - y & \text{if } x > y \\ 0 & \text{if } x \leq y \end{cases}$ [3 marks]

Answer: Note that $monus(x, y)$ is equal to the predecessor function iterated y times, starting from x . Thus $monus$ is λ -defined by $\text{Monus} \triangleq \lambda x y. y \text{ Pred } x$.

(iv) $eq : \mathbb{N}^2 \rightarrow \mathbb{N}$, where $eq(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$ [2 marks]

Answer: Note that $eq(x, y) = and(monus(x, y), monus(y, x))$. Therefore eq is λ -definable by $\text{Eq} \triangleq \lambda x y. \text{And}(\text{Monus } x y)(\text{Monus } y x)$
